STANDARD OPERATING PROCEDURES

Enterprise Software Deployment & Release Management SOP

Pre-Deployment Verification Gates, Canary Rollouts, Health Metrics, and Rollback

Classification: Internal **Version:** 4.3 **Owner:** Site Reliability Engineering (SRE) **Review Cycle:** Semi-Annual

1. Objective and Scope

This Standard Operating Procedure (SOP) defines the mandatory protocols for releasing, verifying, and rolling back production software across Apex Global Technologies. All backend services, frontend portals, and database schemas deployed to our Amazon EKS (Kubernetes) production clusters must adhere to this procedure.

2. Mandatory Pre-Deployment Verification Checklist

Core Query Solution: What steps must be completed before deploying a new application version?

Before initiating any production deployment, the deployment owner must complete and verify the following **seven pre-deployment prerequisites**:

1. **Automated CI Pipeline Pass:** The target commit on `main` must pass all GitHub Actions automated workflows with a 100% test pass rate and minimum 85% code coverage.
2. **Security and Vulnerability Scans:** Automated static code analysis (SonarQube) and dependency scanning (Snyk / pip-audit) must report zero Critical or High severity CVEs.
3. **Staging Environment Verification:** The build artifact (Docker image with immutable SHA256 digest) must be deployed to the Staging environment and pass end-to-end integration and automated smoke test suites.
4. **Change Management & CAB Approval:** A Jira Change Request (RFC) ticket must be logged. For Tier 1 critical services, approval from the Change Advisory Board (CAB) is mandatory.
5. **Database Migration Dry-Run:** Database migrations must be tested against a recent staging snapshot and verified as backward-compatible (two-phase expand/contract schema pattern).
6. **Deployment Window & On-Call Notification:** Standard releases must occur Tuesday through Thursday between 10:00 and 14:00 UTC (Friday and weekend releases are prohibited). The on-call SRE must be notified in Slack `#ops-releases`.
7. **Documented Rollback Plan:** The deployer must confirm the previous stable container image tag and have the rollback command staged.

3. Deployment Pipeline Execution (Canary Rollout)

Deployments execute via ArgoCD using progressive canary rollouts:

```
# Step 1: Deploy canary release (5% traffic allocation)
kubectl apply -f k8s/production/canary-release.yaml

# Step 2: Observe canary health metrics for 15 minutes:
# - HTTP 5xx error rate < 0.05%
# - p99 latency < 250ms
# - Pod restart count == 0

# Step 3: Progressive promotion (25% -> 50% -> 100%)
kubectl argo rollouts promote api-service-rollout
```

4. Automated Rollback Triggers and Manual Rollback Procedure

An automated rollback triggers immediately if error rates exceed 0.5% or latency exceeds 500ms for 3 consecutive minutes. In the event of manual abort:

```
# Emergency Rollback Command
kubectl rollout undo deployment/api-service -n production
kubectl rollout status deployment/api-service -n production
```